

Update Notice

As previously described, duplicate trial entries arising from repeated runs within the same recording session were identified and removed. These duplicates were strictly confined to within-session repetitions and did not occur across different sessions or participants. Following this cleanup, the updated dataset comprises 23,382 trials (corresponding to 93,528 one-second windows). Following duplicate removal, Table 5 presents the fully updated subject-wise sample counts before and after the AutoReject procedure.

Table 5 (Section “Data collection and preprocessing”) Subject-wise 1-second sample counts before and after the AutoReject procedure.

Subject	Trials	Samples before	Samples after	Removed	Removed (%)
sub-01	1,560	6,240	4,918	1,322	21.19
sub-02	1,600	6,400	5,467	933	14.58
sub-03	2,820	11,280	9,610	1,670	14.80
sub-04	1,600	6,400	5,936	464	7.25
sub-05	1,600	6,400	5,273	1,127	17.61
sub-06	1,600	6,400	5,472	928	14.50
sub-07	1,600	6,400	5,373	1,027	16.05
sub-08	1,600	6,400	5,202	1,198	18.72
sub-09	1,600	6,400	6,163	237	3.70
sub-10	1,600	6,400	6,066	334	5.22
sub-11	1,600	6,400	6,152	248	3.88
sub-12	1,600	6,400	6,002	398	6.22
sub-13	1,600	6,400	6,238	162	2.53
sub-14	1,402	5,608	5,056	552	9.84
Total	23,382	93,528	82,928	10,600	11.33

Consequently, all affected experiments were re-evaluated, and the revised results are presented below. Based on our re-executed experiments using the corrected dataset, the performance of EEGConformer in Table 6 improves, whereas the performance of IFNet, ArjunVIT, and ATCNet decreases. Because ATCNet was selected for evaluating the two re-referencing methods in Table 4, Table 4 has been updated accordingly. Furthermore, Figures 8 and 9—which serve as detailed expansions of Table 6—have both been updated as they include these four methods. Please note that all other methods in these tables and figures, as well as the remaining experimental results in the paper, have been verified to remain consistent with the original findings. Among the updated results, EEGConformer achieved the highest cross-trial accuracy of 70.57%, replacing ATCNet as the best-performing method reported in the original article. The cross-session classification and FMA score regression results were not affected by the removal of duplicate trials and are therefore unchanged.

Table 4 (Section “Data collection and preprocessing”) Comparison of MI-classification performance under two reference schemes. [†] indicates results changed from the previous report.

Model	Reference	Accuracy (mean $\pm$ SD, %) [†]	F1 [†]	Kappa [†]
ATCNet [†]	CAR	67.78 $\pm$ 2.57	0.581	0.223
ATCNet [†]	Average mastoid	65.91 $\pm$ 2.74	0.567	0.198
EEGNet	CAR	61.93 $\pm$ 3.18	0.600	0.238
EEGNet	Average mastoid	61.21 $\pm$ 3.06	0.593	0.224

Table 6 (Section “Technical Validation”) Performance comparison of eight EEG decoding models under the 5-fold cross-trial data-splitting strategy. [†] indicates results changed from the previous report; bold indicates the best performance.

Classifier	Accuracy (mean $\pm$ SD, %) $\uparrow$	F1 $\uparrow$	Kappa $\uparrow$
TSLDA + DGFMDRM	53.01 $\pm$ 2.13	0.512	0.060
FBCSP + SVM	56.32 $\pm$ 4.48	0.561	0.126
TWFB + DGFMDRM	56.91 $\pm$ 3.37	0.567	0.138
EEGNet	61.93 $\pm$ 3.18	0.600	0.238
EEGConformer[†]	70.57 $\pm$ 2.58	0.602	0.266
IFNet [†]	67.75 $\pm$ 2.98	0.594	0.235
ArjunViT [†]	64.92 $\pm$ 2.02	0.511	0.123
ATCNet [†]	67.78 $\pm$ 2.57	0.581	0.223

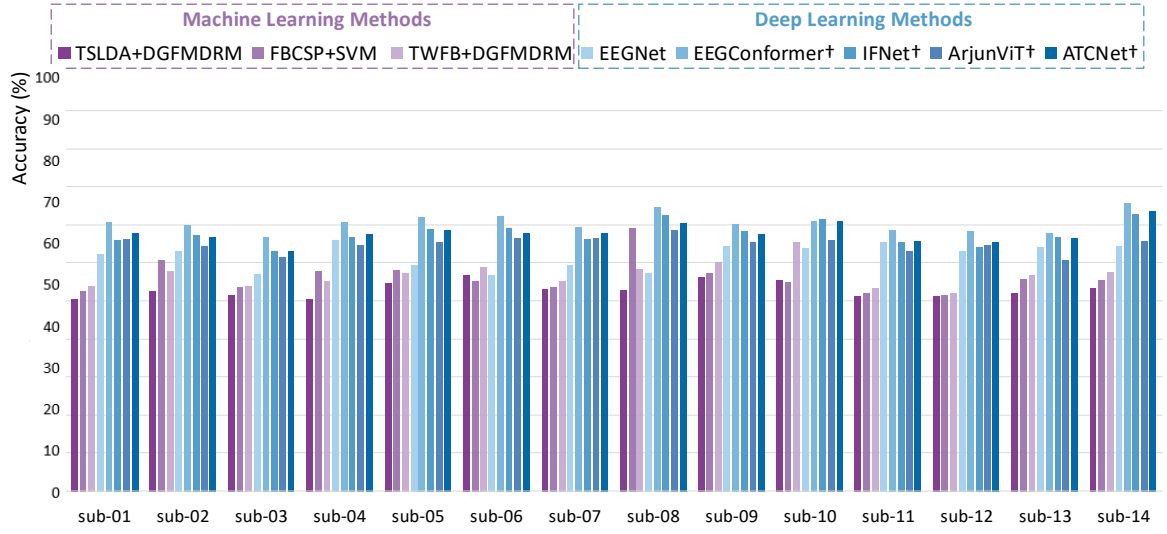


Fig. 8 (Section “Technical Validation”) Performance comparison of eight EEG decoding models across 14 subjects under 5-fold cross-trial data-splitting strategy. [†] indicates results changed from the previous report.

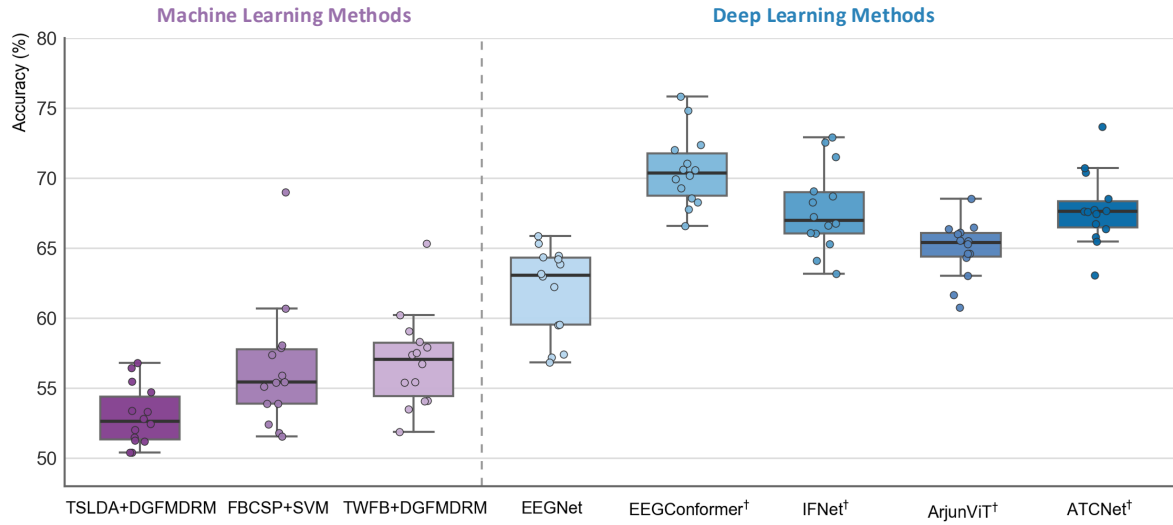


Fig. 9 (Section “Technical Validation”) Box-plot comparison of EEG decoding models across 14 subjects under 5-fold cross-trial data-splitting strategy. [†] indicates results changed from the previous report.